



Head Trauma

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by HandTevy.
- Fluids and Medication titrated to maintain a SBP >70 + (age in years x 2) mmHg
- Hypotension: (SBP < 70+ 2x Age in years)

Signs & Symptoms:

- Altered Mental Status
 - Acute GCS change
- Neuro-deficits
 - Unequal pupils
 - Vision changes
 - Unilateral weakness
 - Nausea/Vomiting
- External head trauma
- Seizures
- Posturing
- Headache

Differential:

- Respiratory failure
- Hypovolemia
- Trauma
- Hypothermia
- Toxins or Overdose
- Hypoglycemia
- Acidosis
- Syncope

Clinical Management Options

P	P	P	P	P	P	• Control external hemorrhage • Evaluate for spinal motion restriction – Do not transport laying on long spine board • Assess GCS score • Elevate patient's head 30° • Administer oxygen to targeted SpO₂ ≥ 92% • Airway management & ventilation COG • Declare Trauma Alert / Activation
L	L	L	L	L	L	• ETCO₂ assessment
1	2	3	4	5	6	• Acetaminophen (all routes) (IV only if SBP ≥ 110) • Vascular access • Adult: Isotonic Crystalloid IV bolus titrated to SBP ≥ 110; Pediatric 20 mL/kg as needed • Tranexamic Acid (TXA) if considering hemorrhage • Advanced airway management • Pain management – Fentanyl and/or Ketamine as indicated. • Push-dose Epinephrine • Blood Product Transfusion and Calcium Chloride to SBP ≥ 110 if hypotension due to hemorrhage • Norepinephrine to SBP ≥ 110 if no external hemorrhage or hemorrhage controlled • RSI Procedure

Consult Online Medical Control As Needed

Inclusion Criteria:

Anyone with physical trauma and a mechanism consistent with the potential to have induced brain injury and any one of the following:

- Loss of consciousness or confusion
- Multisystem trauma requiring intubation
- Post-traumatic seizures

Pearls:

- Minimize Scene time. Interventions should be performed enroute unless extending scene time will have a significant impact on improving patient outcome:
 - Airway management
 - Blood products
- Providers should make all efforts to maximize use of time for best patient outcome.
- Consider H bombs in head trauma: **Hypotension, Hypoxia, Hyperventilation (Avoid these)**
 - Target SBP ≥110
 - Target SPO₂ ≥94%
 - Target EtCO₂ 35-45
- A decrease in mental status may be an early sign of shock.
- Hypotension is devastating to neurologic injury and should be aggressively treated.



Head Trauma

GCS

Eyes Open	Best Verbal	Best Motor
4 – Eyes Open	5 – Oriented	6 – Obeys Commands
3 – To Voice	4 – Confused	5 – Localizes Pain
2 – To Pain	3 – Inappropriate	4 – Withdraws from Pain
1 – None	2 – Incomprehensible	3 – Pain-Flexion
	1 – None	2 – Pain-Extended
		1 – None